SCANFISH

BEHAVIOUR AND TROUBLESHOOTING

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1 Introduction

The aim of this document is to give the ScanFish user a few guidelines to troubleshoot frequently seen issues associated with the operation of the ScanFish. Issues during operation of the ScanFish are often a combination of factors and it can be challenging to isolate the problem area. This document does not claim to provide a solution for all issues but aims to help you to understand and analyse behaviours during operation to handle some of the commonly seen challenges.

2 Behaviour and use of the ScanFish

2.1 Speed of the ScanFish

The recommended survey speed for the ScanFish is normally 4–5 kts speed through water (STW). This may vary depending on the ScanFish configuration. To give the ScanFish the most ideal conditions for maintaining the target depth, the STW should be kept constant; varying speeds will make it hard for the system to follow the target line.

2.2 Cable configuration

The ScanFish can operate while maintaining any depth within the specified range. To ensure stable and reliable ScanFish performance, the cable length needs to be adjusted to match operating depth and speed.

To find the suggested cable length configuration for different survey depth and speed, check the ScanFish cable configuration manual on FreshDesk.

2.3 Flap movements versus ScanFish movement

The height and roll stability are controlled by the port and starboard flaps of the ScanFish. It is possible to monitor the flap angle in ScanFish Flight in the **Data Chart and Numeric Panel**. It is highly recommended to monitor these while operating, as they can tell an experienced operator if something is not behaving as expected.

The flaps can move 40º ±2º in either direction. The centre position is 0º. A positive angle is when the flaps point downwards and the ScanFish descends, while for negative values the flaps point up and the ScanFish ascends.

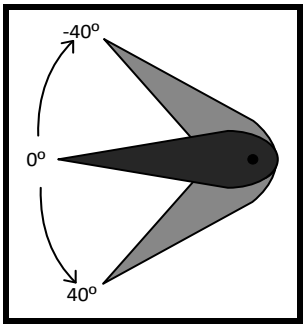

Figure 1 Flap movement showing positive and negative deflection

When the ScanFish ascends or descends, the flaps move simultaneously in the same direction.

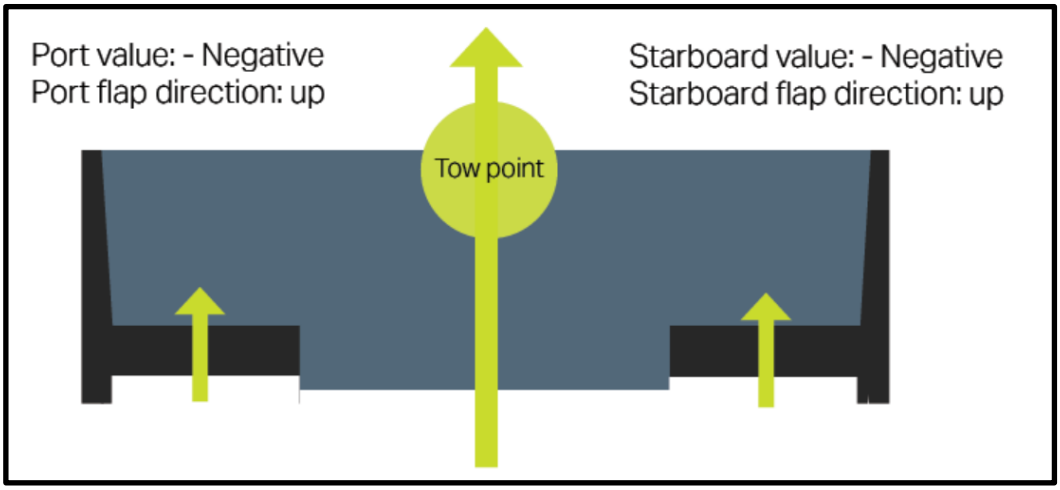

Figure 2 ScanFish ascending

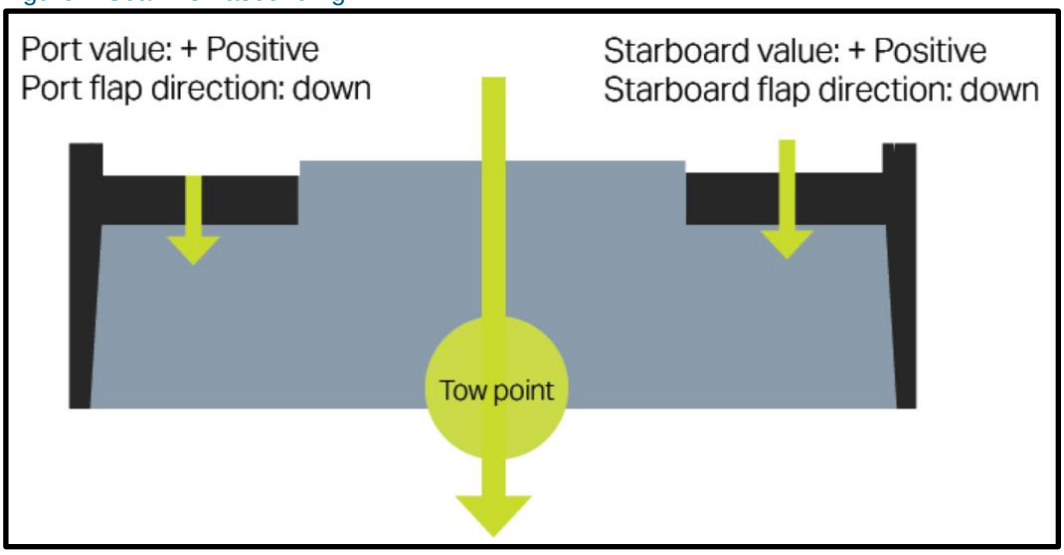

Figure 3 ScanFish descending

When compensating for roll, the flaps move simultaneously in opposing directions

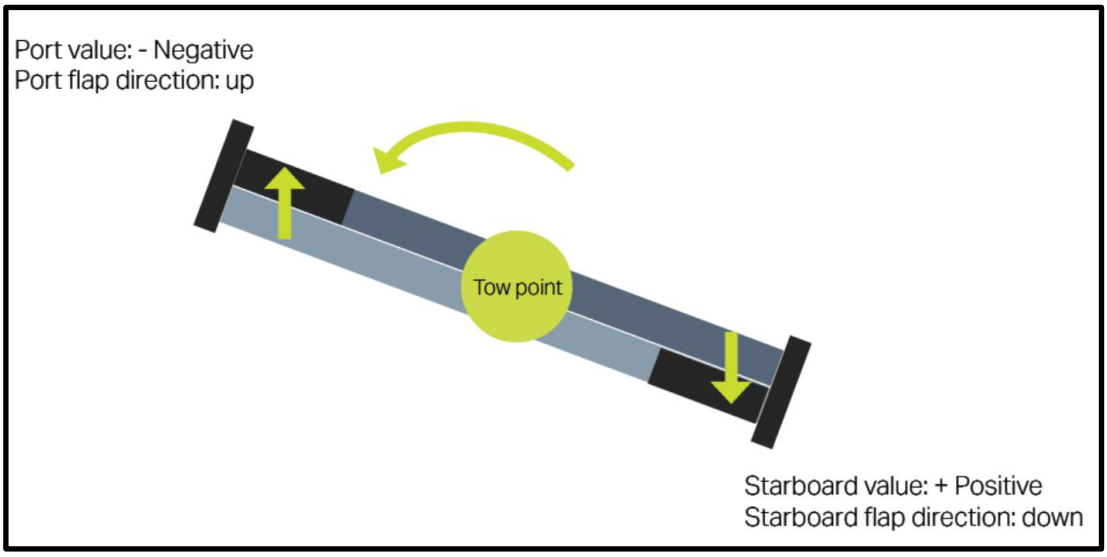

Figure 4 ScanFish compensating for starboard roll

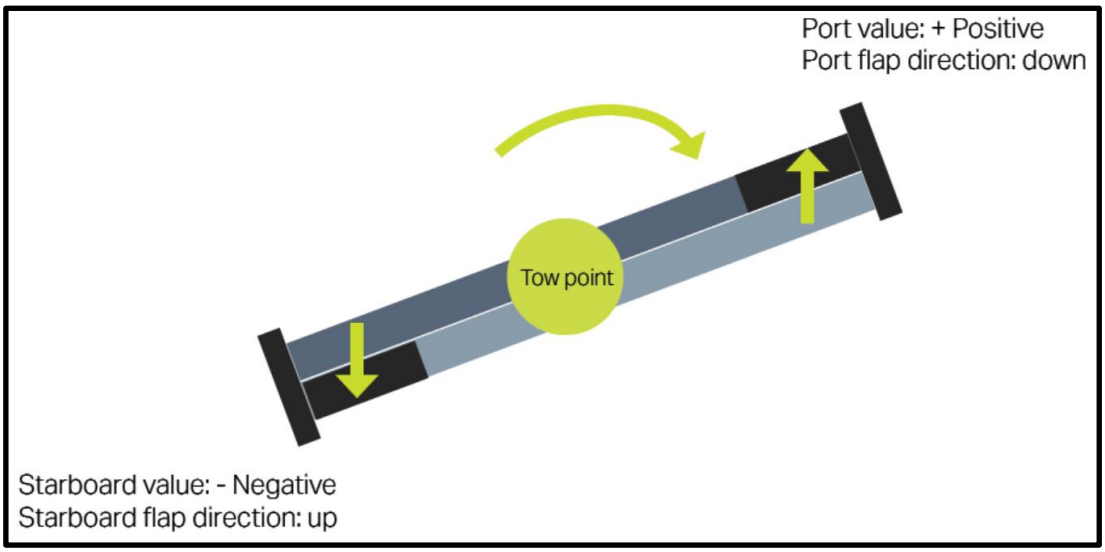

Figure 5 ScanFish compensating for port roll

During operation, the flap movements are a result of motion compensation (roll stabilisation in particular) and the ascending/descending movements from desired target depth. It takes an experienced operator to distinguish the differences, but as a rule thumb, you have a stable, well-performing system when both 2D flap angles are close to neutral. However, it is

not uncommon to have a well-performing system with an offset on the flap angle, as it depends on speed and configuration.

2.4 Balance of the ScanFish

It is important that the ScanFish is physically symmetrical around its centreline. If there is more drag on one side of the centreline compared to the other, the ScanFish will start to roll.

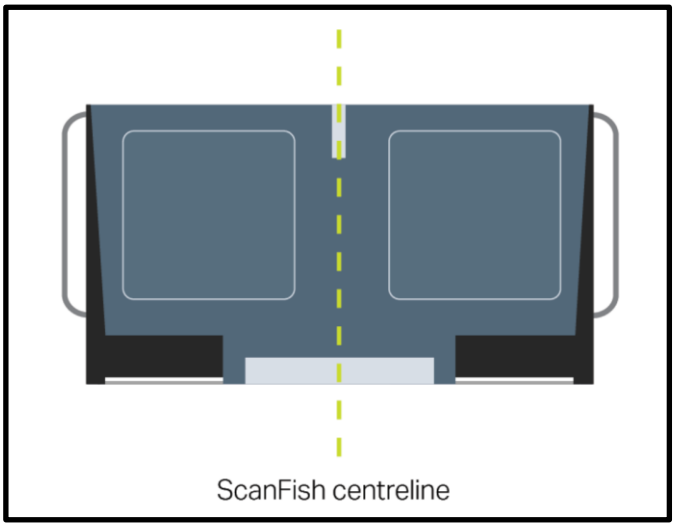

Figure 6 ScanFish centreline

Consequently, never mount an external sensor on one side of the ScanFish, as it will create uneven drag and induce roll in the ScanFish.

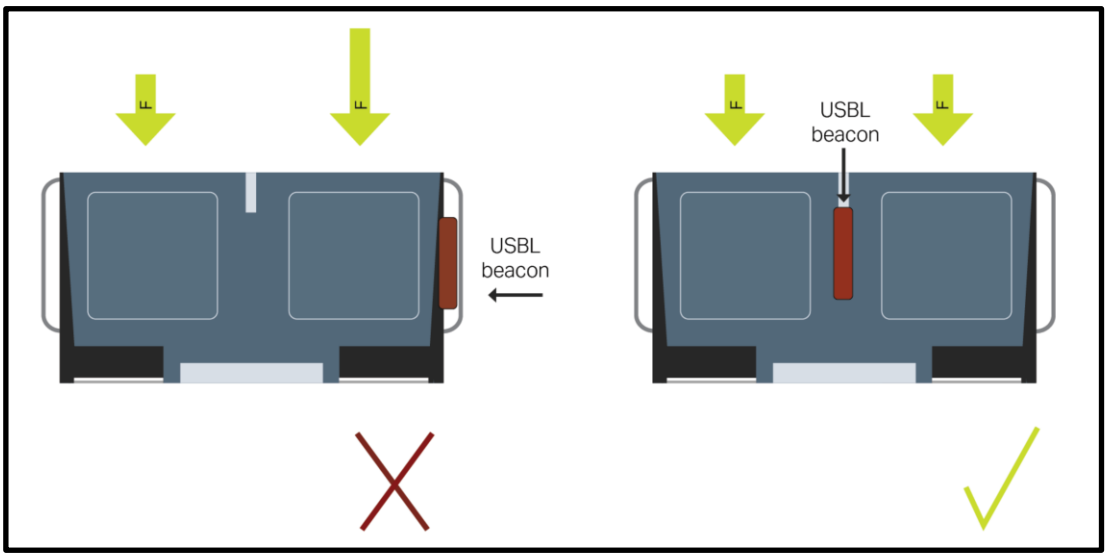

Figure 7 Incorrect versus correct mounting of USBL beacon

On a ScanFish Katria system, it is important to check the cable length on the magnetometers; they must have the same length, or it will introduce an uneven drag and result in roll.

It is also important to check the wear on the fairings. If fairings are damaged, they can create a larger drag and cause the ScanFish to roll.

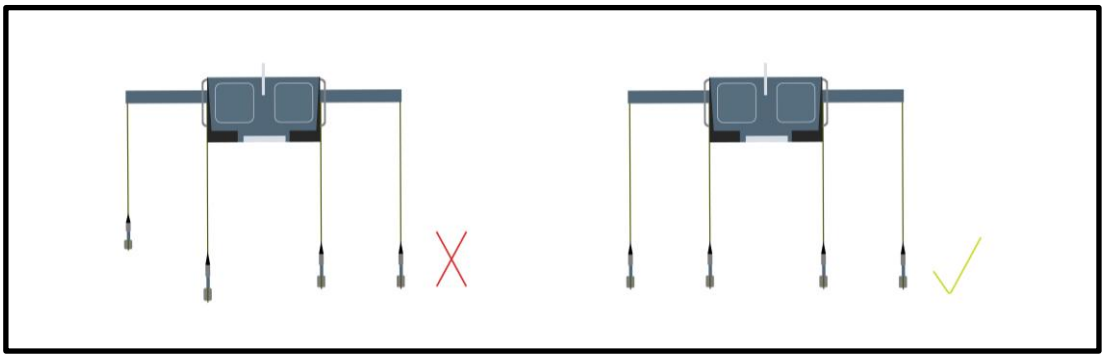

Figure 8 ScanFish Katria

It is possible to get an idea about the balance of the ScanFish system by conducting a live test.

1. Launch the system where you have 20 m of depth or more
2. Sail with or against the current. Set the system speed to ~ 4 kts STW
3. Set the ScanFish to fixed depth 10 m and let the system settle
4. Sail a straight line for 15 minutes while monitoring roll and flap angles

The roll should be < 2º and the average flap angle should be the same for both port and starboard flaps. For example, if the port flap has an average angle of 15º and the starboard 30º it indicates an unbalanced system where there is a larger drag on starboard side.

Note: This test is not conclusive, as there could also be some other external forces such as cable twist, vortex currents, etc, acting on the system. However, it offers an opportunity to check for general balance.

3 Troubleshooting

3.1 The ScanFish cannot track target line

There can be various reason for the ScanFish not being able to track a target depth.

One cause can be the length of the winch cable, which has an influence on the tracking ability.

Too much cable will drag the ScanFish down. The lower the survey speed (STW), the more pronounced the behaviour will be.

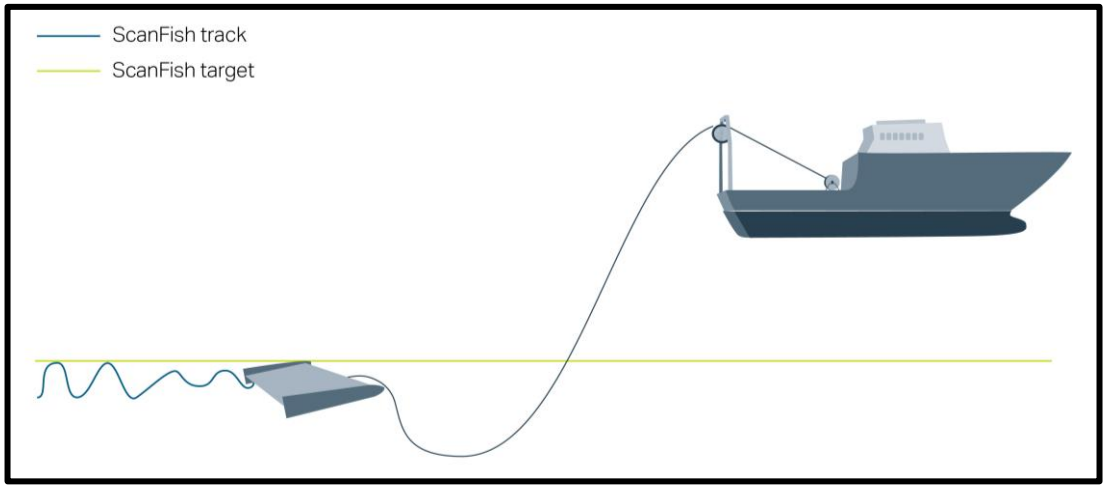

Figure 9 Too much winch cable will drag the ScanFish down

Too little cable will make the ScanFish struggle to get down to the target line and the ScanFish will be more sensitive to surface motion; the ship pitching in the waves will pull the ScanFish toward the surface.

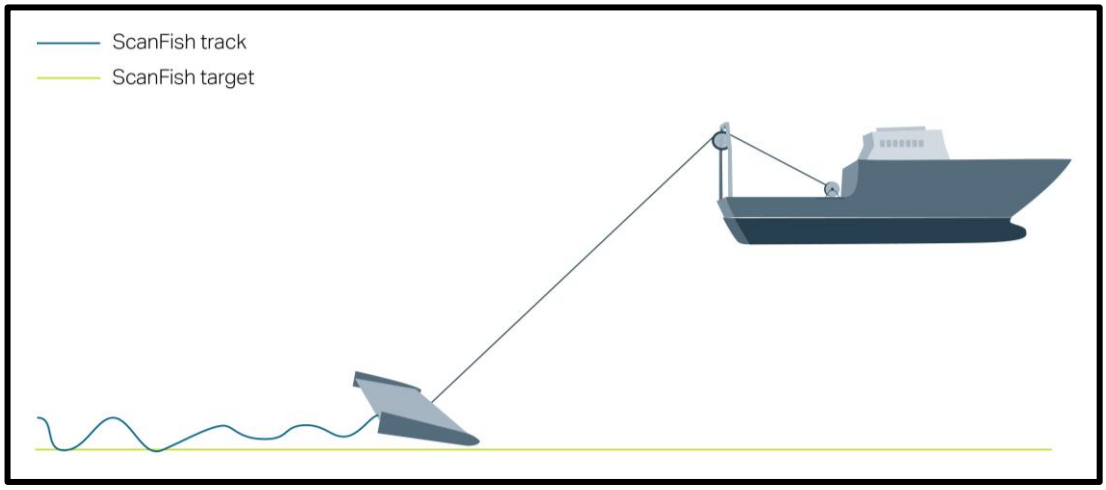

Figure 10 Too little winch cable will make the ScanFish struggle to get to the target depth and be affected by surface movement

3.2 ScanFish is over/undershooting the target line

If the speed of the ScanFish is not constant it will give an unstable tracking of the target depth, resulting in a lot of overshooting or undershooting.

If the speed is slowly increased or decreased, the impact on the ScanFish is very small to nothing. Sudden changes (eg winching in or out) can make the ScanFish overshoot or undershoot its target depth.

When an increase of speed has occurred and the ScanFish starts to overshoot or undershoot, find a constant vessel speed and let the ScanFish settle. Do not speed up and down to make the ScanFish settle, it will only reinforce the overshooting and prolong the settling time.

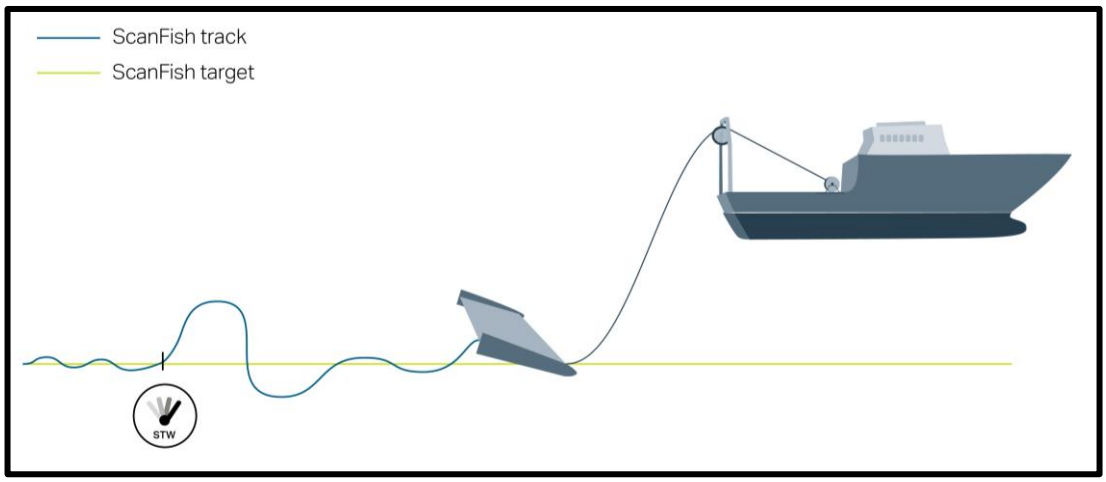

Figure 11 Sudden increase in speed

Similarly, a decrease in speed will make the ScanFish lose lift and make it slowly lose height, until the cable straightens and the ScanFish speeds up and start to overshoot – this movement repeats until a constant speed is maintained and the ScanFish settles.

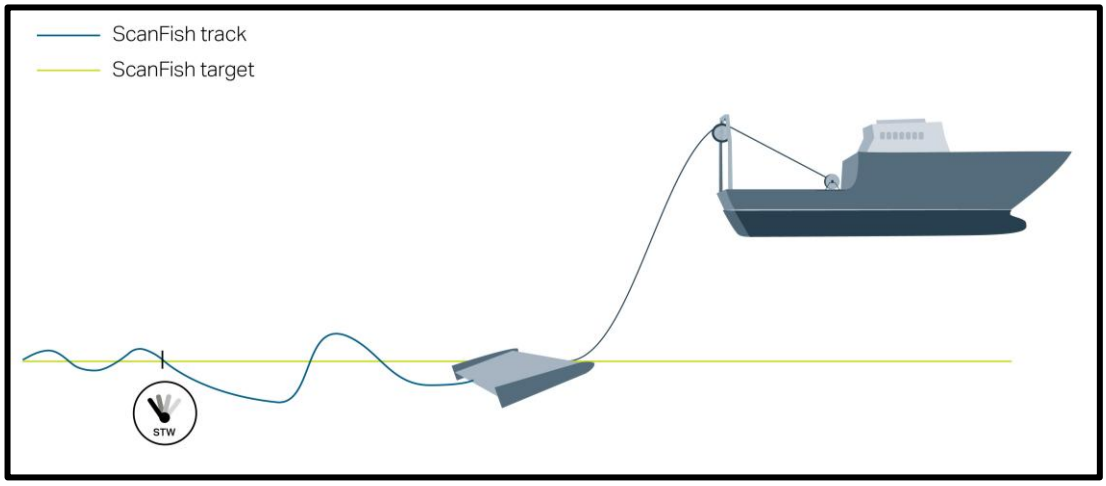

Figure 12 Sudden decrease in speed

3.3 The ScanFish cannot keep the roll within ±2°

While there could be multiple reasons for this, here are several possible causes and solutions to this issue.

3.3.1 One or both flaps may be defective

When the ScanFish is recovered to deck, perform a **Preflight check** in ScanFish Flight. If the check is successful, check for excessive play of the flaps while the ScanFish is still powered on. The flaps should be firm and not show excessive play.

Figure 13 ScanFish flap play check

3.3.2 Equipment may not be symmetrically installed on the ScanFish

Ensure any external sensors/equipment are symmetrically aligned on the ScanFish. For the ScanFish Katria soft-towed solution, ensure the magnetometer cables are equally long and the fairing shows no indication of wear and can rotate freely. The USBL beacon should be placed centrally in the dedicated bracket (Section 2.4).

3.3.3 Flaps are not moving freely due to issues with the panels

Check the presence of the washers between the side panel and the rear plate assembly. The flaps should be able to rotate freely without touching the side panel. If the side panel has been bent due to overtightening of the bottom bolt, the resulting friction between the side panel and flap will leave marks in the side panel and result in roll of the ScanFish.

3.3.4 Roll is occurring under autopilot

If roll starts when autopilot is engaged, limit the flap angle allowed for the autopilot and make the autopilot less aggressive.